



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO TPT Fuel Services

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

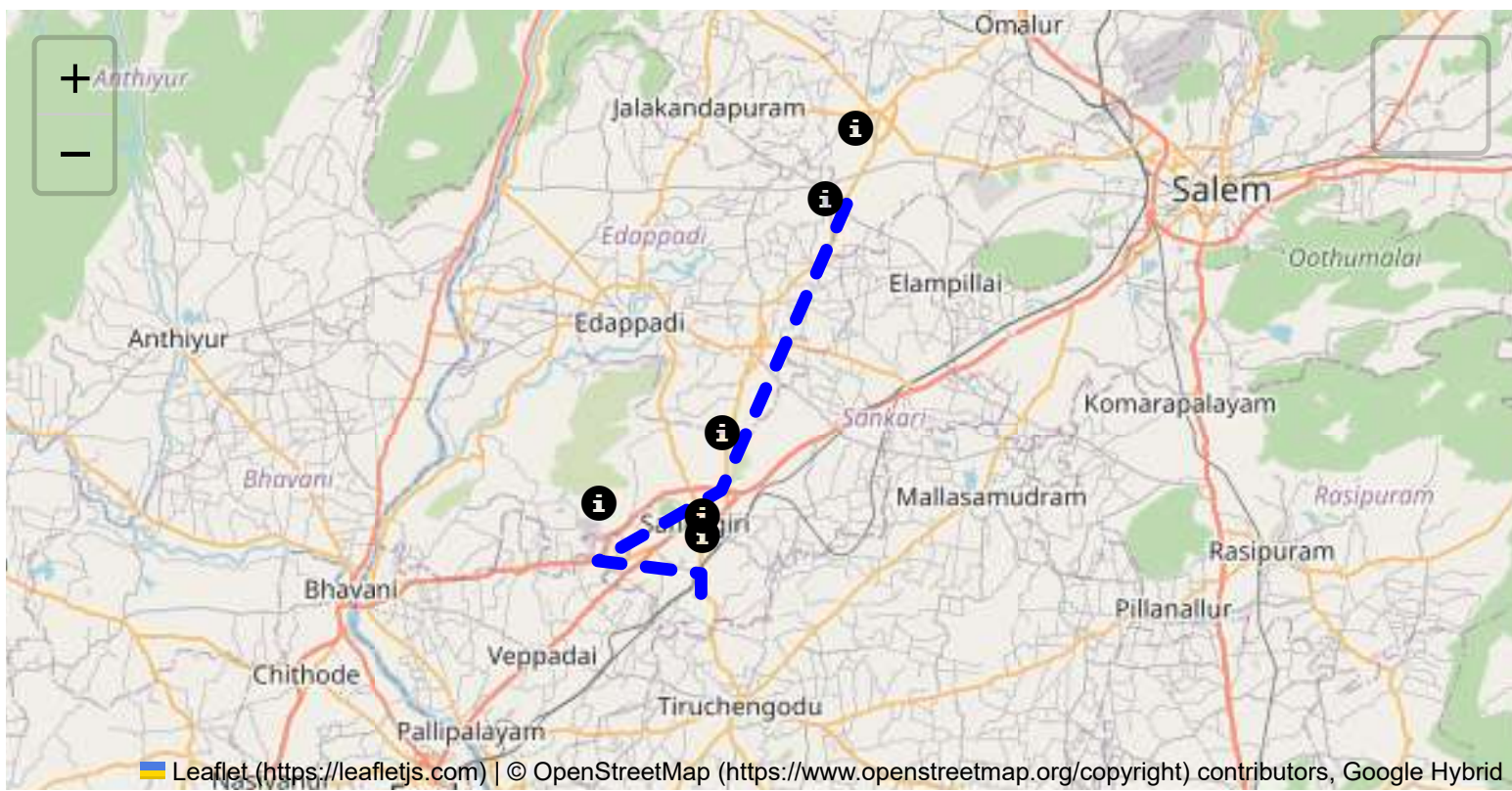
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

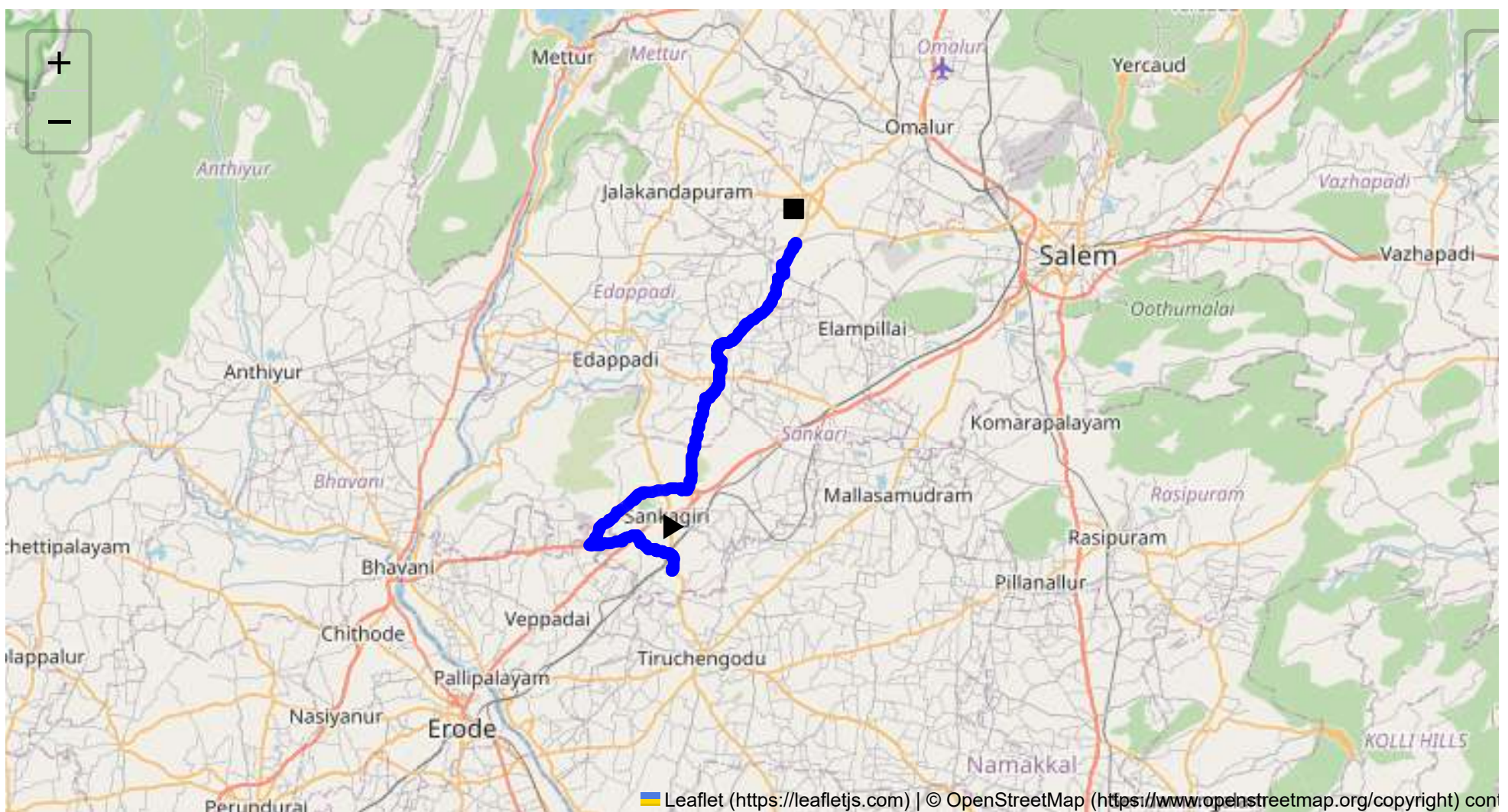
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.65 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.6562503, 77.9569811)



Welcome to the Journey Risk Management Study

- Overview of the Route Map:** The route from CVQF+23W, Sangagiri to MX44+9PQ, SH 86 includes several notable waypoints: CVXF+HPR, FR48+7XQ, FVVM+MCQ, and SH 86 in Koranampatti. The journey spans approximately 40.65 km through a mix of rural roads and secondary highways. This route navigates through small towns and requires careful attention to road signs and local traffic norms.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** The region experiences a tropical climate with distinct dry and monsoon seasons. From June to September, heavy rains can lead to slick roads and localized flooding. January to March are typically dry with moderate temperatures, posing fewer weather-related threats. During the monsoon, drivers should be cautious of slippery roads and reduced visibility.
3. **Analysis of Traffic Patterns:** Traffic congestion can peak during early morning (7:30 AM - 9:30 AM) and late afternoon (4:30 PM - 6:30 PM) due to local commuter traffic. Traffic is generally heavier near Sangagiri and at key intersections in smaller towns. Weekends may see increased traffic volumes due to local agriculture markets and commerce.
4. **Assessment of Road Quality and Infrastructure:** The main thoroughfares are generally well-paved with occasional potholes and rough patches, especially in rural areas. Signage is present but may be less clear in villages. Watch for narrow bridge crossings and unmarked construction areas.
5. **Suggestions for Alternative Routes:** In case of emergencies or roadblocks, one potential detour is via Salem - Namakkal Hwy, which is a longer route but offers better connectivity and road conditions. Consult local traffic advisories for real-time updates.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu imposes restrictions on the transport of hazardous materials during peak daytime hours in urban areas to minimize risk. Compliance with signage about truck routes and regulatory check posts is essential.
7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** There have been isolated incidents of heavy vehicle rollovers due to sharp turns and inadequate awareness of local road conditions. Reports suggest spills due to improper securing of materials. Engage with local transport authorities for the latest safety statistics.
8. **Environmental Considerations and Sensitive Areas:** The route passes near several agricultural zones that can be environmentally sensitive. Drivers should avoid emissions or leaks which could damage these areas. Follow strict protocols for hazardous material transport to avoid contamination.
9. **Analysis of Communication Coverage:** Cell reception is mostly reliable but can drop in rural expanses or dense forested sections. Pre-trip communication checks are advised, and considering backup communication devices like satellite phones or CB radios might be prudent.
10. **Estimated Emergency Response Times:** Emergency response varies, with urban centers like Sangagiri seeing response times of 20-30 minutes. In rural stretches, it could extend to 45-60 minutes. It's important to be aware of the nearest hospitals and fire stations along the route.
11. **Overall Summary of Risk Assessment:** The route presents moderate risks mainly due to weather conditions, rural road infrastructure, and traffic patterns. Careful attention to vehicle maintenance, load security, and compliance with regulations is advised to mitigate these risks. Real-time weather and traffic monitoring, along with predefined emergency response strategies, significantly optimize safety. The route is feasible for hazardous material transport with appropriate precautions, highlighting the need for vigilant driving and contingency planning.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.6624013, 77.9603388	10 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	High	11.49383, 77.88442	15 KM/Hr
15	Turn	Medium	11.49419, 77.88454	30 KM/Hr
16	Turn	High	11.49448, 77.88490	15 KM/Hr
17	Turn	High	11.55614, 77.89764	15 KM/Hr
18	Turn	High	11.58320, 77.90646	15 KM/Hr
19	Turn	Medium	11.64197, 77.95197	30 KM/Hr
20	Turn	Medium	11.64721, 77.95118	30 KM/Hr
21	Blind Spot	Blind Spot	11.66240, 77.96034	10 KM/Hr
22	Turn	High	11.66236, 77.96040	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6624013, 77.9603388



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49419, 77.88015



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88454



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49448, 77.88490



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58320, 77.90646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.66240, 77.96034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66236, 77.96040
